

PRIOR ART DEFENSIVE PUBLICATION

Title: Wearable-Based Detection of Emotional State, Anxiety Disorders, and Dementia-Related Neuropsychiatric Change

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Abstract:

This document discloses systems and methods for detecting emotional arousal, anxiety states (acute and chronic), mood change, and neuropsychiatric manifestations associated with dementia and memory-care populations using wearable physiological sensors. The purpose of this publication is to establish prior art and ensure freedom to operate for researchers and implementers.

Disclosed Approaches Include (Non-Exhaustive):

- Use of heart rate and heart rate variability to infer autonomic arousal and anxiety states.
- Use of electrodermal activity to detect sympathetic activation and panic-related signatures.
- Use of respiratory rate variability and irregular breathing as anxiety indicators.
- Use of sleep latency, fragmentation, and circadian drift as markers of mood and cognitive instability.
- Use of motion/restlessness patterns to detect agitation and dementia-related behavioral disturbance.
- Personalized baseline modeling and adaptive thresholds for individualized detection.
- Longitudinal change detection including anomaly detection, control charts, Bayesian change-point detection, and drift scoring.
- Acute episode detection and panic forecasting via anomaly recognition in resting physiology.
- Differentiation of anxiety versus exertion or illness via contextual activity gating and confound modeling.
- Dementia-specific variants including agitation detection, sundowning window identification, and wandering risk proxies.
- Multimodal fusion variants combining wearable signals with ambient, mobile, or speech-derived features.
- On-device feature extraction and privacy-preserving implementations including federated learning and differential privacy.

Defensive Statement:

All combinations, permutations, obvious extensions, and implementations of the above-described methods for wearable-based emotional, anxiety, mood, and dementia-related detection are intentionally disclosed as prior art. This includes all machine-learning, statistical, rule-based, or hybrid modeling approaches applied to physiological time-series data for mental-health-related inference or change detection.

This disclosure is intended to prevent the patenting of generalized methods described herein and to ensure open research, clinical exploration, and engineering development without fear of patent litigation.